

VIEREMA KAARAKKALA, Finland

WMO: 028340

Lat: 63.841N

Lon: 27.221E

Elev: 208

StdP: 98.86

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-21.1	-25.9	0.4	-23.6	-23.0	0.5	-20.9	8.3	-7.4	7.4	-6.1	2.4	100	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	26.0	17.8	23.9	16.9	22.0	15.7	19.3	23.9	18.1	22.2	16.9	20.5	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.9	20.9	16.5	12.0	19.9	15.2	11.1	18.6	55.9	23.9	51.7	22.3	48.2	20.5	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.4	5.9	DB	-24.9	27.9	2.6	3.1	-26.8	30.1	-28.4	31.9	-29.9	33.7	-31.8	36.0	(4)
(5)			WB	-24.9	20.7	3.6	1.7	-27.4	21.9	-29.5	22.9	-31.5	23.9	-34.1	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-10.2	-7.8	-3.9	2.1	9.0	12.8	16.3	14.2	9.6	2.4	-1.7	-5.6	(6)	
	DBStd	9.88	6.72	6.11	4.39	3.69	5.04	4.21	3.81	3.40	3.40	3.61	4.18	5.77	(7)	
	HDD10.0	3015	626	498	432	239	78	18	2	6	47	236	351	483	(8)	
	HDD18.3	5576	884	731	690	488	291	172	87	135	263	493	601	741	(9)	
	CDD10.0	520	0	0	0	2	48	103	197	135	34	1	0	0	(10)	
	CDD18.3	38	0	0	0	0	2	7	23	7	0	0	0	0	(11)	
	CDH23.3	225	0	0	0	0	18	41	135	32	0	0	0	0	(12)	
	CDH26.7	27	0	0	0	0	1	3	21	3	0	0	0	0	(13)	
	WSAvg	3.1	3.2	3.5	3.5	3.2	3.2	3.0	2.5	2.8	2.8	3.1	3.4	3.6	(14)	
(15) Precipitation	PrecAvg	702	48	40	36	35	56	74	89	77	64	63	60	59	(15)	
	PrecMax	949	94	80	69	60	112	153	158	156	106	108	118	136	(16)	
	PrecMin	519	13	4	3	12	14	27	21	26	21	12	10	18	(17)	
	PrecStd	109	21	20	16	13	26	32	36	29	25	27	26	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	4.1	7.2	17.0	25.6	26.8	28.6	26.5	19.9	11.7	8.4	4.0	(19)	
		MCWB	1.9	2.5	3.2	9.4	14.3	17.5	19.2	19.4	15.5	10.0	7.8	3.9	(20)	
	2%	DB	0.5	2.1	4.9	12.3	22.7	24.4	26.7	23.7	17.3	10.0	5.7	2.5	(21)	
		MCWB	0.3	1.3	1.9	6.7	13.5	16.6	18.8	17.3	13.6	8.9	5.4	2.2	(22)	
	5%	DB	-0.1	0.7	2.9	10.1	20.2	21.9	24.6	21.6	15.6	8.6	4.3	1.3	(23)	
		MCWB	-0.2	0.3	1.0	5.6	12.6	15.0	18.0	16.1	12.8	7.8	4.1	1.1	(24)	
	10%	DB	-1.0	-0.1	1.5	7.9	17.2	19.8	22.5	19.6	14.2	7.3	3.1	0.4	(25)	
		MCWB	-1.1	-0.3	0.3	4.2	10.7	13.9	16.7	15.3	12.1	6.6	3.0	0.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.9	3.8	10.0	16.0	19.3	21.2	20.4	15.8	11.0	7.8	3.9	(27)	
		MCDB	2.1	3.9	6.3	14.5	22.6	23.6	25.6	24.9	19.2	11.6	8.2	4.0	(28)	
	2%	WB	0.4	1.5	2.5	7.8	14.5	17.7	19.8	18.1	14.6	9.3	5.5	2.3	(29)	
		MCDB	0.4	2.1	4.1	11.2	20.7	22.1	24.5	22.1	16.5	9.7	5.8	2.4	(30)	
	5%	WB	-0.1	0.4	1.5	5.9	13.0	15.9	18.8	16.9	13.4	8.0	4.1	1.1	(31)	
		MCDB	-0.1	0.7	2.6	9.2	19.6	21.0	23.3	20.2	14.9	8.5	4.3	1.3	(32)	
	10%	WB	-1.1	-0.3	0.5	4.5	11.3	14.6	17.7	16.0	12.6	6.7	3.0	0.3	(33)	
		MCDB	-1.0	-0.1	1.3	7.2	16.8	18.9	21.6	18.9	13.9	7.3	3.1	0.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.8	6.1	6.9	8.4	8.0	7.8	7.2	5.9	4.3	3.5	3.9	(35)	
		MCDBR	3.3	4.6	6.9	10.4	11.4	10.3	9.6	9.2	7.8	5.7	4.0	3.6	(36)	
	5% WB	MCWBR	3.4	4.2	4.9	6.2	5.7	4.9	4.4	4.3	4.7	4.6	4.1	3.5	(37)	
		MCWBR	3.3	4.4	5.6	9.5	10.6	9.1	8.8	8.0	6.8	5.3	3.9	3.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.213	0.276	0.303	0.341	0.340	0.340	0.363	0.345	0.323	0.302	0.367	N/A	(40)	
		taud	1.976	2.124	2.184	2.303	2.463	2.481	2.432	2.471	2.525	2.396	2.697	N/A	(41)	
	Edn at Noon		368	634	780	835	875	880	844	829	778	622	345	N/A	(42)	
		Edh at Noon	28	63	90	101	95	95	98	86	66	49	25	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.16	0.69	2.14	3.58	4.74	5.25	4.98	3.76	2.08	0.78	0.19	0.06	(44)	
	RadStd		0.02	0.15	0.23	0.42	0.52	0.38	0.52	0.38	0.22	0.13	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page